Manufacturing of semiconductors is done for the whole world.

"When you build a system you have a microprocessor that will require very state-of-the-art packaging versus other parts that will require different types of packaging. Whatever we do we should always intend to be a global player," says Raja. He adds, "Hence getting into the ATMP/OSAT story is a great beginning for us."

Harshad Mehta of Visicon reveals the tidbits about the call he had with MeitY and how the Ministry was of firm view that power semiconductors is the low hanging fruit. The point that differentiates a power semiconductor fab from most other fabs is the fact that a power semiconductor fab requires investment of around $50 to $60 million, which when compared to other fabs is a much lower investment. Moreover, power semiconductors cater to strategic sectors like aerospace and defense.

"The margins are high, there is not much competition, and the power semiconductors are quick in terms of delivering RoI. Power Electronics is a sector where vertical integration is the key," says Harshad.

Big question: Investment and jobs

India today enjoys the largest amount of FDI. The government has been launching one PLI scheme after another promising billions in the field of electronics manufacturing, and now the talk of India having its own semiconductor ecosystem is gaining momentum once again. But what's important is whether this semiconductor ecosystem will be able to create more jobs in the country, and how much in terms of investment will it be able to attract?

"Just consider the scale of consumption of electronic components in the country and calculate roughly what that consumption will look like in the year 2025. If we were to produce semiconductors for India to consume in the country (minus the memory), it would require us to establish at least 30 fabs. That is approximately a $120 billion investment. Can we afford such investments in the next five years? Obviously not," answers Krishna.

Countries including the United States, Taiwan, and Korea have been talking about $200 billion worth of investments for the semiconductor ecosystem during the last three months. It simply means that the world will see a lot more semiconductors being manufactured. Now the question is where will these millions of semiconductors get tested and packaged? The answer, as Krishna pointed out is, "That is the opportunity India is talking about."

The ideal time for a wafer to come out from the fab and getting ready for shipment is usually 36 hours. Anything more than 36 hours is seen as the ATMP unit losing the game. To enable this time window, the government and the private players will need to work on setting up an ideal infrastructure.

"This infrastructure consists of not only the ATMP units but warehouses at airports, bigger airports, excellent transport facilities, seamless custom clearance abilities, and a lot more. These areas will also help other sectors, will create more jobs, and also attract investment," says Krishna. He adds, "If we are able to develop this kind of infrastructure, I guarantee that billions of dollars' worth of investment will come to India."

In terms of jobs, the challenge does not lie in finding individuals in India but in training people for the right skills. Even 12th pass students can be trained on how to operate machines and equipment at OSATs and ATMPs. That, as per Krishna, is the beauty of having an OSAT.

"The best part is governments are willing to foot the bill. They are willing to have their people skilled for the technologies of the future," notes Krishna. As a matter of fact, IESA is working on introducing a skilling programme around electronics and semiconductors in the country. The association has plans to open the programme for both engineering as well as below engineering students.

"It is a great time for Indians working on semiconductors outside India to look back at the country. Raja and I have been working on semiconductors for more than 30 years and it took us no time to jump on a plane to come back to India," says Venkata.

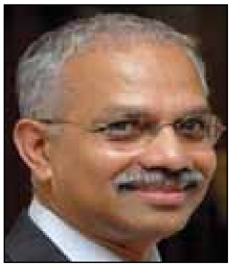
He adds, "It is also a good time to start focusing on ITIs and polytechnics and start helping students for the next wave of modernisation that is set to take place in the country." EFY

Speakers

Dr Harshad Mehta, head, Silicon Power

Krishna Moorthy KM, CEO, IESA

Poornima Shenoy, former president, IESA, and CEO of Hummingbird Advisors



Raja Manickam, CEO, OSAT Venture, Tata

Venkata Simhadri, founder, ASIP Tech, and CEO of MosChip

Mukul Yudhveer Singh is a business editor at EFY